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A Project Proposal
on
“altEco”

[Code No: COMP 207]

(For partial fulfillment of II Year/ II Semester in Computer Engineering)

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Submission Date: 2025/05/09

Abstract

With the increasing global emphasis on environmental sustainability, consumers are seeking reliable ways to make eco-conscious purchasing decisions. However, challenges such as greenwashing, inconsistent labeling, and limited access to trustworthy environmental data hinder the identification of genuinely sustainable products. This project proposes the development of a dedicated platform that leverages artificial intelligence to recommend eco-friendly consumer products based on user preferences and standardized sustainability metrics. The system will analyze diverse data points-including carbon footprint, material sourcing, manufacturing practices, and eco-certifications-to generate personalized recommendations and assign sustainability scores to products. Additional features include educational content, impact tracking, and community-driven insights to promote informed and responsible consumption. By offering accessible, actionable information and alternatives, the platform aims to empower users to reduce their environmental footprint, foster behavioral change toward sustainable lifestyles, and support the broader goals of waste reduction, resource conservation, and decreased carbon emissions.

Keywords: *Sustainability, Artificial Intelligence, Greenwashing, Carbon footprint, Data Analysis*

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Acronyms/Abbreviations

- AI: Artificial Intelligence
- API: Application Programming Interface
- CSS: Cascading Style Sheets (as in Tailwind CSS)
- DB: Database (implied in context, e.g., MongoDB)
- GPU: Graphics Processing Unit
- IDE: Integrated Development Environment
- JS: JavaScript (as in ReactJS, Node.js, Express.js)
- ML: Machine Learning
- OS: Operating System (as in Windows, macOS, Linux)
- RAM: Random Access Memory
- SSD: Solid State Drive
- UI: User Interface

Chapter 1 Introduction

1.1 Background

With growing awareness of the environmental impact of consumer habits, individuals are increasingly motivated to make eco-conscious decisions in their daily purchases. (Ashley Reichheld, 2023) The push for sustainability has led to a rising demand for products that are ethically sourced, biodegradable, recyclable, or have a low carbon footprint. However, despite this demand, consumers often face difficulty identifying genuinely sustainable products due to greenwashing, inconsistent labeling, and limited access to reliable environmental data. (§enyapar, 2024)

Most current e-commerce platforms and product search tools are not equipped to provide detailed insights into the sustainability of the products they feature. While some platforms highlight eco-tags or offer generic filters like “organic” or “vegan,” these features lack depth and consistency. As a result, consumers are left without a comprehensive way to compare the environmental impact of their choices or to discover more responsible alternatives that align with their values. (Emmanuel Mireku, 2024)

This project proposes a dedicated platform that leverages artificial intelligence to assist users in making more sustainable shopping choices. By analyzing a wide range of data points—such as carbon footprint, material sourcing, manufacturing practices, and eco-certifications—the system will generate personalized recommendations for eco-friendly products. The platform will also offer educational content, impact tracking features, and community-driven insights, helping users become more informed and proactive in reducing their environmental footprint. Through this approach, the solution aims to empower everyday consumers to contribute meaningfully to sustainability, not just through awareness but through measurable actions.

1.2 Objectives

The major objectives of altEco are listed below:

- i.To recommend eco-friendly consumer products based on user preferences and sustainability metrics.
- ii.To assign and display sustainability scores for products using standardized environmental criteria.
- iii.To allow customization of recommendations based on ethical and ecological filters.
- iv.To track users' environmental impact through eco-friendly product choices.

1.3 Motivation and Significance

In today's world, environmental sustainability has become a global priority, yet many individuals remain unaware of the harmful impacts of everyday products they use. (Sousa, 2023) The project __ is driven by the need to provide accessible and practical information that helps users identify commonly used objects and discover their eco-friendly alternatives. By offering clear, sustainable options, the project encourages more responsible consumption habits and supports a transition toward greener lifestyles. This initiative is significant not only in promoting individual awareness but also in contributing to broader environmental goals such as waste reduction, resource conservation, and decreased carbon footprint.

1.4 Expected Outcomes

- i.A working system that takes user input and provides eco-friendly alternatives to common objects.
- ii.Enhanced public awareness and knowledge of sustainable choices.
Behavioral change toward environmentally responsible consumption.
- iii.Support for eco-friendly product markets through increased visibility and usage.

Chapter 2 Related Works

Some existing platforms related to our project are mentioned below:

2.1 Yuka

Yuka (Yuka, 2025) is a mobile app that helps users make healthier and safer choices by evaluating food and cosmetic products. It uses a rule-based scoring system to assess nutritional content, additives, and the risk level of ingredients, assigning products a score from 0 to 100 and provides recommendations for healthier or safer alternatives. Based on this score, Yuka recommends healthier alternatives but does not personalize recommendations or focus on environmental sustainability. While Yuka is highly effective in helping users make informed decisions about their health, it primarily focuses on nutrition and ingredient safety.

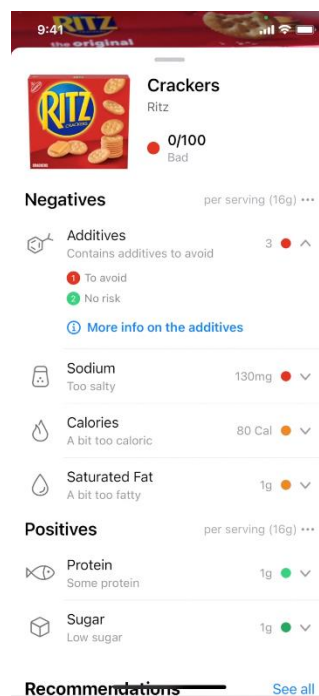


Figure i: Yuka App

2.2 Green Choice

Green Choice (Green Choice, 2025) is a mobile application designed to assist users with their food choices. They not only take into consideration the nutritional value of various options but also how eco-friendly they are. Elements such as organic certification and carbon footprint are taken into account. This analysis yields a Green Score, which forms the basis of product recommendations, providing readily accessible replacements with better sustainability profiles. While Green Choice encourages conscious consumption with a broad overview of product impacts, its recommendations are primarily based on the overall Green Score and general product attributes, providing generalized suggestions rather than personalized ones tailored to specific dietary needs or values beyond its core sustainability framework.

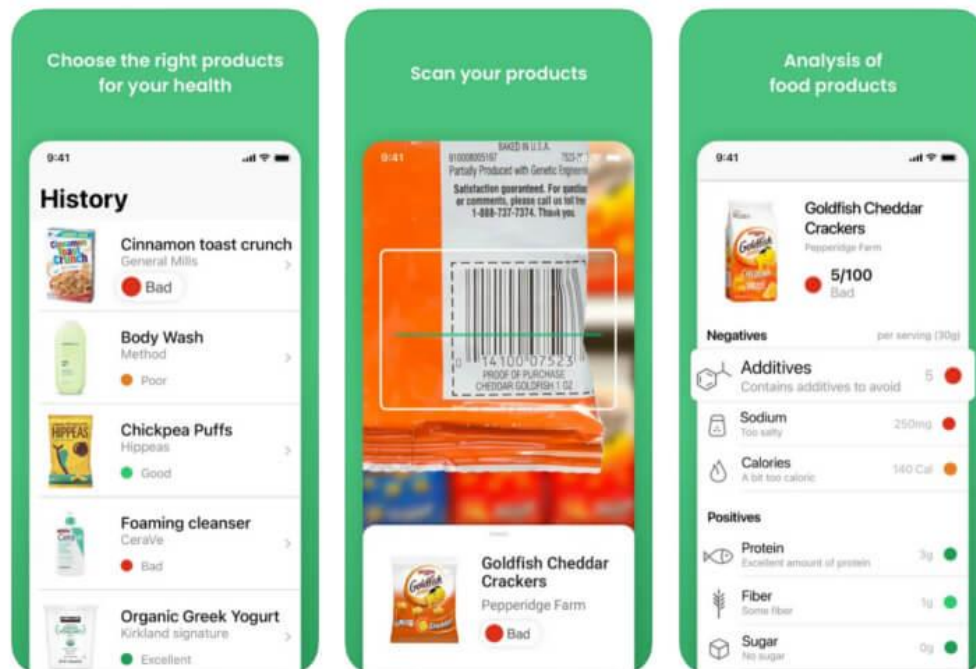


Figure ii: Green Choice App

Chapter 3 Procedure and Methods

The procedure that will be followed in the course of development of altEco is systematically presented below:

i. Research and Study:

This initial phase involves examining existing eco-friendly product platforms and recommendation systems. The team will identify limitations in current solutions and explore the latest technologies suitable for development. This phase also includes studying sustainability metrics like carbon footprint and exploring machine learning models for personalized recommendations. Work distribution among team members will also be finalized here.

ii. Front-End Design:

This phase emphasizes the development of a visually appealing, responsive and intuitive user interface using ReactJS. The goal is to ensure a seamless user experience across all devices, with accessibility and usability as core principles. React Router will be utilized for dynamic and efficient client-side routing, enabling smooth transitions between different sections. Tailwind CSS will be leveraged to implement a clean and modern visual design that adapts well to both desktop and mobile devices. The UI will emphasize simplicity and clarity, ensuring users can easily interact with machine learning-driven suggestions and explore environmentally responsible alternatives.

iii. Core Programming:

This development phase involves implementing core functionality such as user registration, login, preference management and product recommendation workflows. The backend will be built using Node.js and Express.js, with MongoDB serving as the database to store product data, user preferences, eco-scores and

impact tracking data. A machine learning model will analyze product information and generate personalized eco-friendly recommendations.

iv. Testing and Debugging:

Comprehensive testing will be conducted to ensure platform stability and correctness. Unit testing and manual testing will be used to verify each component. The team will actively debug issues, address performance bottlenecks and ensure the app meets security and privacy standards, especially in user data handling.

v. Documentation:

The final phase includes preparing detailed project documentation and technical reports. This also includes creating presentations and visual summaries of the system architecture, data flow, and ML workflow. Proper documentation ensures smooth handover, understanding and possible future upgrades.

Chapter 4 System Requirement Specification

4.1 Software Specification

- i. Operating System: Windows 10/11, macOS, or any Linux distribution
- ii. Front-End: ReactJS, Tailwind CSS, React Router
- iii. Back-End: Node.js (v16 or above), Express.js
- iv. Database: MongoDB (local or Atlas cloud instance)
- v. Machine Learning: Python (v3.8 or above), Scikit-learn, Pandas, NumPy
- vi. Version Control: Git, GitHub
- vii. API Testing Tools: Postman, Insomnia
- viii. Development Tools: VS Code / WebStorm / any preferred IDE
- ix. Package Manager: npm or yarn
- x. Browser Support: Chrome, Firefox, Safari, Edge

4.2 Hardware Specification

- i. Processor: Intel Core i5 (8th Gen) or equivalent
- ii. Memory: 8 GB of RAM.
- iii. Storage: 256 GB SSD
- iv. Graphics: Dedicated GPU (NVIDIA GTX 1650 or above) for ML training

Chapter 5 Project Planning and Scheduling

The dedication and contribution of each member is required for the completion of this project. Teamwork plays a significant role to achieve this goal. In order to develop the web application in the given time, which is limited, we decided to break it down. It will help us to accomplish the tasks of this project work smoothly and calmly without having to rush at the end, avoiding last-minute stress. The rough estimation of the time allocation for different tasks is illustrated below in the Gantt chart.

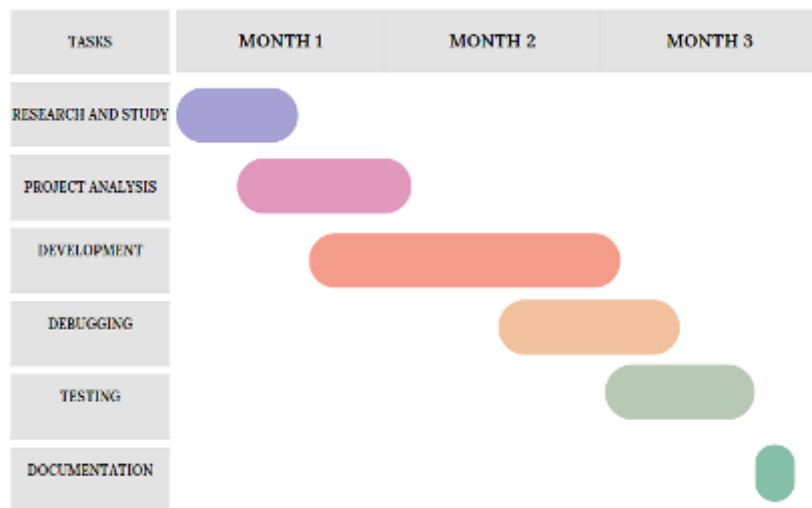


Figure iii: GANTT Chart

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